INCH-POUND

MIL-DTL-83723/90E
15 September 2006
SUPERSEDING
MIL-DTL-83723/90D
20 October 2000

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SOLDER FLANGE MOUNT, THREADED COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, N, AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

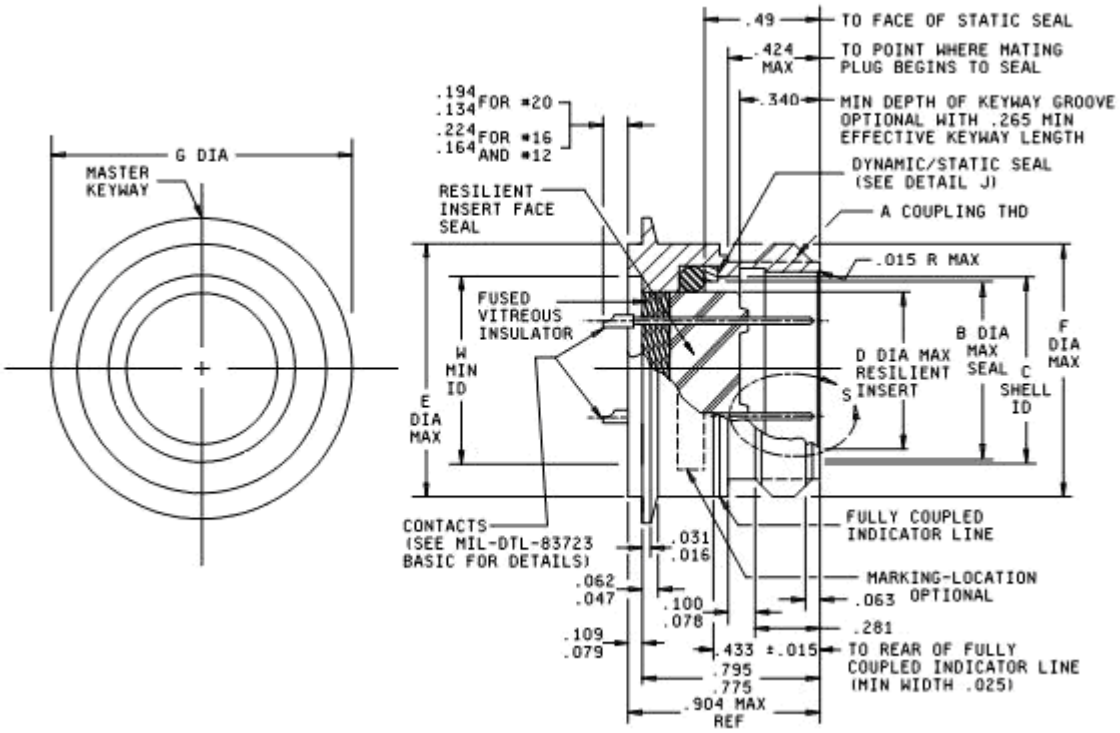



FIGURE 1. Dimensions and configurations.



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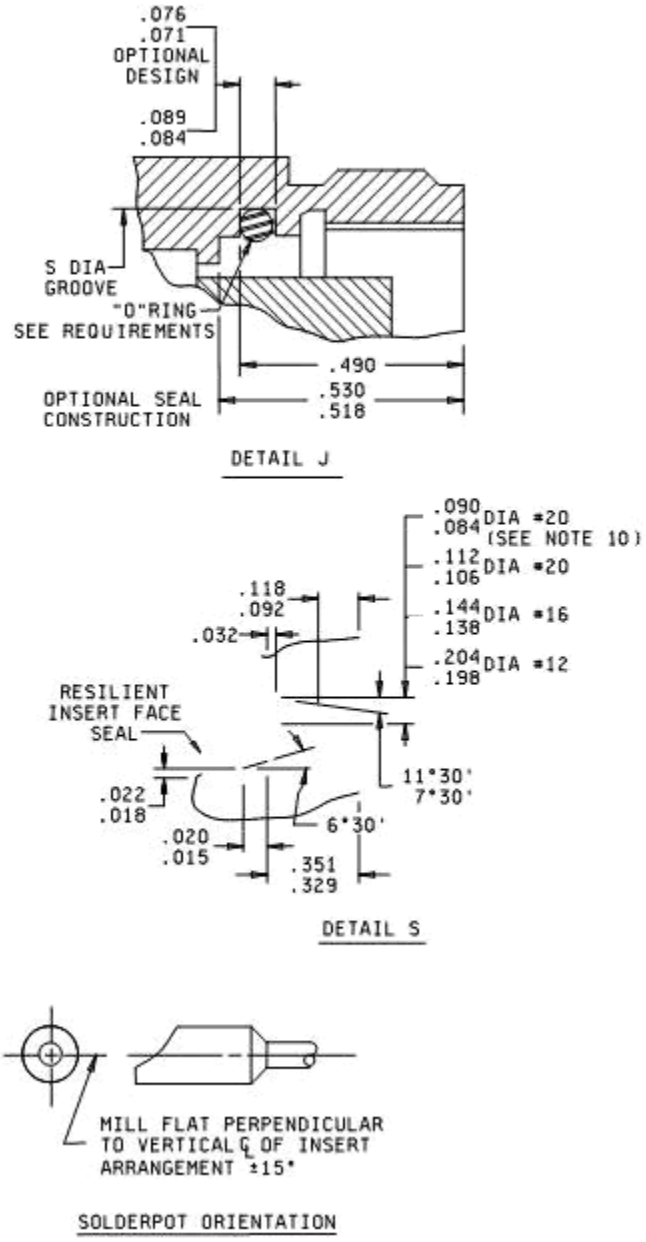



FIGURE 1. Dimensions and configurations - Continued.



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| Shell size | A Thread UNEF-2A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 8 | .562-24 | .394 | .418<br>.413 | .290 | .500<br>.494 | .562 |
| 10 | .688-24 | .515 | .535<br>.530 | .388 | .562<br>.556 | .696 |
| 12 | .875-20 | .685 | .700<br>.705 | .558 | .750<br>.744 | .875 |
| 14 | .938-20 | .754 | .774<br>.769 | .627 | .812<br>.806 | .936 |
| 16 | 1.062-18 | .881 | .901<br>.896 | .772 | .937<br>.931 | 1.062 |
| 18 | 1.188-18 | .987 | 1.007<br>1.002 | .860 | 1.062<br>1.056 | 1.187 |
| 22 | 1.438-18 | 1.237 | 1.257<br>1.252 | 1.110 | 1.312<br>1.306 | 1.437 |

| Shell size | G | W | X | S<br>+.005<br>-.000 |
|---|---|---|---|---|
| 8 | .723<br>.703 | .400 | .510 | .490 |
| 10 | .850<br>.830 | .411 | .572 | .636 |
| 12 | 1.055<br>1.035 | .581 | .760 | .806 |
| 14 | 1.100<br>1.080 | .650 | .822 | .875 |
| 16 | 1.220<br>1.200 | .765 | .947 | 1.002 |
| 18 | 1.350<br>1.330 | .864 | 1.072 | 1.108 |
| 22 | 1.600<br>1.580 | 1.133 | 1.322 | 1.358 |

FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .003 | 0.08 | .112 | 2.84 | .490 | 12.45 | .765 | 19.43 | 1.035 | 26.29 |
| .004 | 0.10 | .118 | 3.00 | .494 | 12.55 | .769 | 19.53 | 1.055 | 26.80 |
| .005 | 0.13 | .134 | 3.40 | .500 | 12.70 | .772 | 19.61 | 1.056 | 26.82 |
| .015 | 0.38 | .138 | 3.51 | .510 | 12.95 | .774 | 19.66 | 1.062 | 26.97 |
| .016 | 0.41 | .144 | 3.66 | .515 | 13.08 | .775 | 19.69 | 1.072 | 27.23 |
| .018 | 0.46 | .148 | 3.76 | .518 | 13.16 | .795 | 20.19 | 1.080 | 27.43 |
| .020 | 0.51 | .160 | 4.07 | .530 | 13.46 | .806 | 20.47 | 1.100 | 27.94 |
| .022 | 0.56 | .164 | 4.17 | .535 | 13.59 | .812 | 20.62 | 1.108 | 28.14 |
| .025 | 0.64 | .194 | 4.93 | .556 | 14.12 | .822 | 20.88 | 1.110 | 28.19 |
| .031 | 0.79 | .198 | 5.03 | .558 | 14.17 | .830 | 21.08 | 1.133 | 28.78 |
| .032 | 0.81 | .204 | 5.18 | .562 | 14.27 | .850 | 21.59 | 1.187 | 30.15 |
| .047 | 1.19 | .224 | 5.69 | .572 | 14.53 | .860 | 21.84 | 1.200 | 30.48 |
| .062 | 1.57 | .265 | 6.73 | .581 | 14.76 | .864 | 21.95 | 1.220 | 30.99 |
| .063 | 1.60 | .281 | 7.14 | .627 | 15.93 | .875 | 22.23 | 1.237 | 31.42 |
| .071 | 1.80 | .290 | 7.37 | .636 | 16.15 | .881 | 22.38 | 1.252 | 31.80 |
| .076 | 1.93 | .329 | 8.36 | .650 | 16.51 | .896 | 22.76 | 1.257 | 31.93 |
| .078 | 1.98 | .340 | 8.64 | .685 | 17.40 | .901 | 22.89 | 1.306 | 33.17 |
| .079 | 2.01 | .351 | 9.92 | .696 | 17.68 | .904 | 22.96 | 1.358 | 34.49 |
| .084 | 2.13 | .388 | 9.86 | .700 | 17.78 | .931 | 23.65 | 1.312 | 33.32 |
| .089 | 2.26 | .394 | 10.01 | .703 | 17.86 | .936 | 23.77 | 1.322 | 33.58 |
| .090 | 2.29 | .400 | 10.16 | .705 | 17.91 | .937 | 23.80 | 1.330 | 33.78 |
| .092 | 2.34 | .411 | 10.44 | .723 | 18.36 | .947 | 24.05 | 1.350 | 34.29 |
| .100 | 2.54 | .413 | 10.49 | .744 | 18.90 | .987 | 25.07 | 1.358 | 34.49 |
| .106 | 2.69 | .418 | 10.62 | .750 | 19.05 | 1.002 | 25.45 | 1.437 | 36.50 |
| .109 | 2.77 | .424 | 10.77 | .754 | 19.15 | 1.007 | 25.58 | 1.580 | 40.13 |
| | | .433 | 11.00 | .760 | 19.30 | 1.022 | 25.96 | 1.600 | 40.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm) for three place decimals and ± .01 (0.25 mm) for two place decimals and ±1° for all angles.
4. Dimensions apply after plating.
5. 63 surface finish ( $^{63}\surd$ ) all over in accordance with ASME-B46.1 unless otherwise specified.
6. Remove all burrs and sharp edges .003 to .005 R.
7. Diameters to be concentric within .004 TIR.
8. Diameter in the same cross-sectional plane to be concentric with each other within .003 TIR.
9. Shell size 8 will not mate with MIL-DTL-26500 connectors.
10. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

This connector mates with MIL-C-83723/86, /91, /97 and MIL-DTL-83723/95.

Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

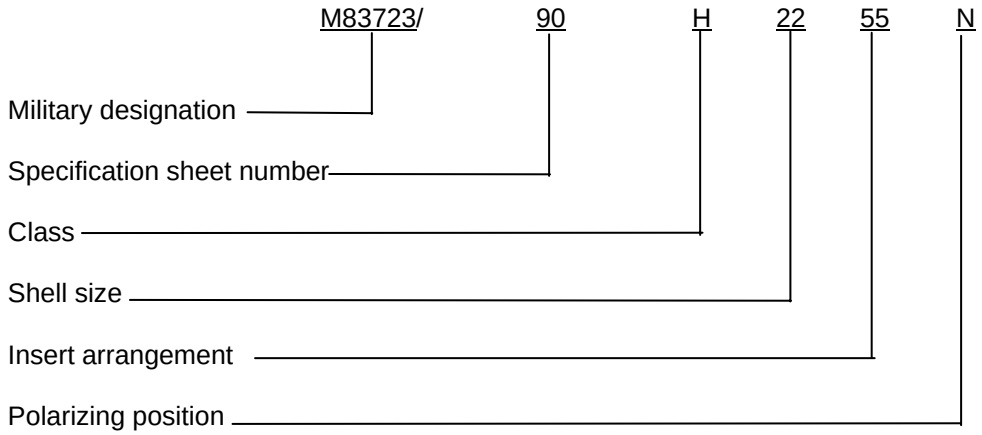
TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

| Shell size | Insert arrangements | Contact cavities |
|---|---|---|
| 8 | 03, 98 | 1, 2, 3 |
| 10 | 06 | 1, 2, 3, 4, 5, 6 |

Connector accessories: See SAE-AS85049.

“O” ring: See MIL-DTL-83723.

Part or Identifying Number (PIN) example:




Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-C-83723/86
MIL-C-83723/91
MIL-C-83723/97
MIL-DTL-26500
MIL-DTL-83723/95
MIL-STD-1554
ASME-B46.1
SAE-AS85049



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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, MI
Navy - EC
Air Force - 19, 99

(Project 5935-2005-063)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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